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KAUST ACADEMY

جامعة الملك عبدالله للعلوم والتقنية
King Abdullah University of Science and Technology

ISSUE #01 ▪ 25 August 2026

Weekly AI Brief

19–25 August 2026

Papers, model releases, benchmarks and policy from the past seven days

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The week in five lines

- **Coding agents cannot yet migrate a repository.** On the new SWE Refactor Bench, 28 of 520 runs (5.4%) passed all three grading stages; the best model, `claude-opus-5`, scored 47.0/100. [S3]
- **DeepSeek put vision into V4-Flash at text prices.** Images bill at up to 384 tokens each on the existing V4-Flash rate card. API only, no weights, and every quoted score is DeepSeek's own. [S1]
- **A harness, not a model, moved ARC-AGI-3.** Prime Agent reports RHAE Best@1 rising from 30% to 95.5% with the code released. [S4]
- **Alibaba raised HK$80bn (US$10.2bn) on 23 August** in a Hong Kong share placement earmarked for full-stack AI capability. [S10]
- **Nvidia is reported to be paying Poolside $6bn** to license its Model Factory, plus $1bn invested at a $12bn pre-money valuation. Nvidia has not announced this. [S11]

1. Model and product releases

DeepSeek-V4-Flash-Vision-Exp

DeepSeek • experimental API model • link • [S1]

DeepSeek added image input to V4-Flash on 21 August. Images are tokenised for billing at up to 384 tokens each and charged at the existing V4-Flash rate card, so vision carries no separate premium. The model is served through the Chat Completions, Messages and Responses APIs and accepts base64, external URLs or a new free Files API; there are no downloadable weights. DeepSeek states that on multimodal agent benchmarks the model "makes a major leap over V4-Flash, bringing multimodal agent performance close to Opus-4.8" — that comparison is self-reported and we found no independent reproduction of it.

Why it matters. Vision at text-token prices removes the usual cost argument for keeping a separate captioning model in an agent pipeline.

Groq 3 LPX in full production; Vera Rubin NVL72

NVIDIA • hardware announcements, 24 August • link • [S2]

NVIDIA announced on 24 August that its Groq 3 LPX inference accelerator has entered full production, and published performance claims for Vera Rubin NVL72 of up to 30× more work per watt on agentic workloads. A companion release states that SpaceXAI will adopt the Vera CPU.

The per-watt figure is a vendor claim on a vendor-chosen workload mix; the newsroom items do not publish the benchmark methodology behind it.

Why it matters. If the per-watt claim survives third-party measurement, inference cost per agent trajectory — not per token — becomes the number worth budgeting against.

2. Research worth reading

Candidates were drawn from arXiv cs.CL new listings and from the community ranking on Hugging Face Daily Papers for 25 August [S12]; every arXiv identifier, date and figure below was then read from the paper's own abstract page rather than from the ranking.

SWE Refactor Bench: Can Coding Agents Complete a Long-Horizon, Whole-Repository Stack Migration?

Hong et al. • arXiv:2608.23564 • link • [S3]

The benchmark asks agents to migrate an entire repository to a different stack, and its central contribution is a grading protocol rather than a task set. The authors name a failure they call "Blindness": agents pass tests by copying the original code instead of migrating it. Their three stages check migration completion, behavioural correctness, and hidden behavioural differences via independent agent verification. Across 8 frontier models and 26 model-effort configurations (520 runs), 28 runs — 5.4% — passed all three stages, and 13 of 20 tasks received no accepted solution at all. The gap by task type is the useful signal: 31.4 on build-toolchain rewrites against 5.6 on language rewrites.

Why it matters. A test suite that goes green is not evidence of a completed migration, which makes independent behavioural checks a prerequisite for trusting agent refactors.

Prime Agent: A Self-Improving RLM Harness

Karten, Zhang, Thomas et al. • arXiv:2608.23552 • link • [S4]

An open-source harness that persists histories, memories, skills, prompts and subagent specifications across trajectories, built around a persistent IPython REPL following the Recursive Language Model abstraction. The reported headline is ARC-AGI-3 RHAE Best@1 moving from 30% to 95.5%, with further results on long-context coding, GPU-kernel generation and autonomous nanoGPT speedruns. Code is released under CC BY 4.0 at `github.com/PrimeIntellect-ai/prime-agent`. The numbers are the authors' own; what makes them checkable is that the harness ships.

Why it matters. It relocates a large capability gain from model weights to scaffolding, which is the cheaper half of the stack to change.

Apodex 1.1: Scaling Agentic Intelligence for Complex Work

Apodex Team (70 authors) • arXiv:2608.23283 • link • [S5]

A report on two scaling axes for agents: Environment Scaling, which broadens the diversity and verifiability of executable file, search and code environments, and Agentic Coordination Scaling, which trains decomposition, delegation of parallel work, integration of asynchronous results and replanning. A shared execution harness and an "AgentOS" hold state across tools. A 35B locally deployable variant, Apodex 1.1 Mini, is described. The paper claims a leading performance band across finance, scientific research, mathematics, coding and search, but the abstract page does not carry the per-benchmark table, and we did not find a weights release to check against.

Why it matters. The claim under test is that environment diversity, not parameter count, is now the binding constraint on agent quality.

On the Threat Model of Weird Generalization and Emergent Misalignment

Wanner, Dredze & Walden • arXiv:2608.23476 • link • [S6]

Narrow fine-tuning on small datasets can produce broad, unintended behavioural change. This paper takes that phenomenon apart and finds it depends far more on dataset composition and language than on dataset size, that it is stronger when the fine-tuning data resembles pretraining data than when it is novel, and that measurements of it are sensitive to which evaluation questions are asked. The authors conclude it behaves as an adversarial threat requiring deliberate data engineering, rather than as a hazard inherent to routine fine-tuning.

Why it matters. It downgrades a widely cited risk from ambient to adversarial, which changes who has to defend against it and how.

MobilePA-Bench: Benchmarking Mobile Planner Agents on Complex Real-World Tasks

Zhu, Wu, Wang et al. • arXiv:2608.23035 • link • [S7]

An interactive benchmark that evaluates mobile agents in an executable sandbox maintaining live application databases, spanning 13 functional domains and 212 tools, rather than scoring screen interactions or API string matches. It probes sub-agent collaboration, memory for implicit requests, and invocation of pre-packaged skills. The authors report that frontier LLMs remain unreliable in this setting, with performance dropping sharply under strict tool ordering, permission limits and unexpected runtime errors.

Why it matters. Stateful evaluation exposes the failure modes — ordering, permissions, runtime errors — that static tool-call benchmarks are structurally blind to.

ReWorld and EchoWM: two interactive world models

Chen et al.; Zhang et al. • arXiv:2608.23565, arXiv:2608.23189 • link • [S8]

ReWorld targets the tension between responsive user control and long-horizon memory, using mixed attention windows with global heads plus a pose-indexed landmark bank on a fixed inference memory budget. It reports 11.95° rotation error, 64-second out-and-back rollouts at 704×1280, and best control fidelity against six recent interactive world models. EchoWM generates interactive 720p video together with environmental sound, music and speech, and unifies discrete and continuous control across first- and third-person scenes through a shared trajectory representation [S9]. Neither abstract page states weights availability.

Why it matters. Persistent scene memory under a fixed budget is what separates a video generator from something an agent can be trained inside.

3. Benchmarks, evaluations and reproductions

Benchmark	Top entry	Score	Note
SWE Refactor Bench	`claude-opus-5`	47.0	Best of 8 frontier models; 5.4% of 520 runs pass all 3 stages
– build-toolchain subset	(all models)	31.4	Task-type split within the same benchmark
– language-rewrite subset	(all models)	5.6	Same models, same harness; the harder half
ARC-AGI-3 (RHAE, Best@1)	Prime Agent harness	95.5	Up from 30.0; harness change, model held constant

Every figure above is reported by the authors of the benchmark or harness that produced it, and none has an independent reproduction as of 25 August. The two are not comparable to each other: SWE Refactor Bench grades whether a migration actually happened across 520 runs, while the ARC-

AGI-3 figure is a Best@1 selection over repeated attempts, which is a materially easier statistic. The SWE Refactor Bench result is the more load-bearing of the two, because its contribution is an audit protocol designed to catch agents that pass tests without doing the work — the failure mode it names is one that previous migration benchmarks would have scored as success [S3]. Treat the ARC-AGI-3 jump as a claim awaiting third-party runs; the code being public makes that check cheap, and it is the single most worthwhile reproduction of the week [S4].

4. Industry, funding and policy

- **Alibaba raised HK$80bn (US$10.2bn).** The Hong Kong share placement was reported on 23 August, with the company stating proceeds would fund full-stack AI capability including expanded AI infrastructure. Share count and price per share were not confirmed against the filing itself for this issue. [S10]
- **Nvidia and Poolside: $6bn licence, $1bn investment.** Newcomer reported on 20 August, citing a letter to investors, that Nvidia will pay $6bn to license Poolside's Model Factory and invest a further $1bn at a $12bn pre-money valuation. This is single-source reporting from a document, not an announcement: it does not appear in Nvidia's newsroom, and neither company has confirmed it. Headcount figures circulating alongside it could not be verified. [S11]
- **The policy beat was quiet.** The month's binding dates — EU AI Act transparency obligations and California SB 942 — both fell on 2 August, outside this window. No new AI regulation, enforcement action or export-control change published between 19 and 25 August could be confirmed against a primary source, so none is reported here.

5. What to watch next week

- Whether anyone reproduces Prime Agent's 30% → 95.5% ARC-AGI-3 RHAE result. The code is public, so an independent run is a weekend's work and the absence of one after two weeks would itself be informative. [S4]
- Whether Nvidia confirms the Poolside arrangement. A $6bn licence should surface in a newsroom post or a quarterly filing; if it does not, the investor-letter sourcing stays the only evidence. [S11]
- Whether DeepSeek promotes V4-Flash-Vision out of "exp" status and publishes a benchmark table others can run. Right now the Opus-4.8 comparison rests entirely on DeepSeek's own measurement. [S1]
- Whether SWE Refactor Bench attracts submissions from labs other than the eight already evaluated. A migration benchmark that only its authors run does not stay a benchmark for long. [S3]

6. Sources

#	Reference
S1	DeepSeek. *DeepSeek-V4-Flash-Vision-Exp release note.* 21 Aug 2026. https://api-docs.deepseek.com/news/news260821
S2	NVIDIA. *Newsroom: Groq 3 LPX full production; Vera Rubin NVL72; SpaceXAI Vera CPU.* 24 Aug 2026. https://nvidianews.nvidia.com/news
S3	Hong, D. et al. *SWE Refactor Bench: Can Coding Agents Complete a Long-Horizon, Whole-Repository Stack Migration?* arXiv:2608.23564, 24 Aug 2026. https://arxiv.org/abs/2608.23564
S4	Karten, S. et al. *Prime Agent: A Self-Improving RLM Harness.* arXiv:2608.23552, 24 Aug 2026. https://arxiv.org/abs/2608.23552
S5	Apodex Team. *Apodex 1.1: Scaling Agentic Intelligence for Complex Work.* arXiv:2608.23283, 24 Aug 2026. https://arxiv.org/abs/2608.23283
S6	Wanner, M., Dredze, M. & Walden, W. *On the Threat Model of Weird Generalization and Emergent Misalignment.* arXiv:2608.23476, 24 Aug 2026. https://arxiv.org/abs/2608.23476
S7	Zhu, Y. et al. *MobilePA-Bench: Benchmarking Mobile Planner Agents on Complex Real-World Tasks.* arXiv:2608.23035, 24 Aug 2026. https://arxiv.org/abs/2608.23035
S8	Chen, Z. et al. *ReWorld: An Interactive World Model with Long-Horizon Memory.* arXiv:2608.23565, 24 Aug 2026. https://arxiv.org/abs/2608.23565
S9	Zhang, S. et al. *EchoWM: Open and Enterable Omnimodal World Models.* arXiv:2608.23189, 24 Aug 2026. https://arxiv.org/abs/2608.23189
S10	South China Morning Post. *Alibaba to issue HK$80 billion in new shares for global AI push.* 23 Aug 2026. https://www.scmp.com/tech/big-tech/article/3364957/alibaba-issue-hk80-billion-new-shares-global-ai-push
S11	Newcomer. *Poolside Strikes $6 Billion Licensing Deal with Nvidia.* 20 Aug 2026. https://www.newcomer.co/p/sources-poolside-strikes-6-billion
S12	Hugging Face. *Daily Papers, 25 August 2026.* https://huggingface.co/papers

Compiled 25 August 2026. Every figure in this brief was read from the linked primary source on that date. Where a number is self-reported or single-sourced, the sentence carrying it says so. Corrections: `Ali Habibullah`.